

# UNIT V — BRYOPHYTA

**Definition — Bryophytes:** Small, non-vascular land plants traditionally grouped as mosses, liverworts and hornworts, characterized by a dominant gametophyte and a multicellular dependent sporophyte.

## 5.1 Introduction and history of bryology

**Definition — Bryology:** The branch of botany dealing with bryophytes.

Bryophytes are important in the study of land-plant evolution because they show adaptation to terrestrial life while still requiring free water for key stages of sexual reproduction. The gametophyte is the conspicuous, independent phase; the sporophyte remains attached to and nutritionally dependent on the gametophyte to varying degrees.

## 5.2 Thallus structure

| Group      | Dominant gametophyte form                      | Examples in syllabus |
|------------|------------------------------------------------|----------------------|
| Liverworts | Thalloid or leafy                              | Riccia, Marchantia   |
| Hornworts  | Thalloid gametophyte with horn-like sporophyte | Anthoceros           |
| Mosses     | Leafy shoot with protonemal stage              | Polytrichum          |

## 5.3 Reproduction

### Vegetative reproduction

- Fragmentation of the thallus or plant body.
- Specialized propagules such as gemmae in suitable taxa.
- Protonemal multiplication in mosses.

### Sexual reproduction

Bryophytes are oogamous. The male sex organ is the **antheridium**, producing motile male gametes; the female sex organ is the **archegonium**, containing the egg. Water is generally required for male gametes to reach the archegonium.

#### General bryophyte sexual reproduction



## 5.4 Life cycle and alternation of generations

**Definition — Alternation of generations:** A life-cycle pattern in which multicellular haploid and multicellular diploid phases occur in sequence.

#### Core alternation of generations in bryophytes



**Definition — Key idea:** The gametophyte is the dominant visible generation. The sporophyte is produced after fertilization and remains attached to the gametophyte.

## 5.5 Riccia

- A thalloid liverwort with dorsiventral thallus.
- Thallus commonly shows dichotomous branching and a median groove.
- Sex organs are embedded in the thallus.
- The sporophyte is highly reduced in the classical description.

## 5.6 Marchantia

- Dorsiventral, dichotomously branched thallus.
- Upper surface bears air pores leading to photosynthetic chambers.
- Ventral surface bears rhizoids and scales.
- Vegetative reproduction by gemmae in gemma cups is a key diagram.
- Sexual structures are borne on specialized stalked receptacles in the classical account.

Vegetative reproduction by gemmae in Marchantia



## 5.7 Anthoceros

- Thalloid hornwort.
- Cells commonly contain a large chloroplast with a pyrenoid in the classical description.
- Sporophyte is elongated and horn-like.
- Persistent basal meristem allows continued sporophyte growth for a period.

## 5.8 Polytrichum

- Representative moss with leafy gametophytic shoots.
- Life cycle includes a protonemal stage followed by leafy gametophores.
- Sporophyte includes foot, seta and capsule.
- Capsule structure and spore release are important morphological study points.

| Feature             | Riccia             | Marchantia                         | Anthoceros           | Polytrichum               |
|---------------------|--------------------|------------------------------------|----------------------|---------------------------|
| Major group         | Liverwort          | Liverwort                          | Hornwort             | Moss                      |
| Gametophyte         | Thalloid           | Thalloid                           | Thalloid             | Leafy                     |
| Distinctive feature | Reduced sporophyte | Gemma cups and stalked receptacles | Horn-like sporophyte | Protonema and leafy shoot |

## 5.9 Economic and ecological importance

- Pioneer colonization of bare substrates and participation in ecological succession.
- Water retention and soil-surface stabilization in suitable habitats.
- Peat-forming moss systems are ecologically important carbon stores.
- Bryophytes can serve as indicators of environmental conditions because of their direct interaction with water and surrounding air.

**Exam focus:** Unit V must-master diagrams: Marchantia thallus with gemma cups, alternation of generations, Anthoceros horn-like sporophyte and Polytrichum moss plant/capsule.